Auteur: Rob Willemsen
Studierichting: Informatica
Oefeningenreeks 3C nr. 17.6

$$
\begin{aligned}
&\lim_{x\to 0}\frac{\sin x - x\cos x}{x(1-\cos x)} = \frac{0}{0}\\
&\stackrel{H}{=} \lim_{x\to 0}\frac{\cos x - \cos x + x\sin x}{1-(\cos x + x\sin x)}\\
&= \lim_{x\to 0}\frac{x\sin x}{1-\cos x + x\sin x} = \frac{0}{0}\\
&\stackrel{H}{=} \lim_{x\to 0}\frac{\sin x + x\cos x}{\sin x + \sin x + x\cos x}\\
&= \lim_{x\to 0}\frac{\sin x + x\cos x}{2\sin x + x\cos x} = \frac{0}{0}\\
&\stackrel{H}{=} \lim_{x\to 0}\frac{\cos x + \cos x - x\sin x}{2\cos x + \cos x - x\sin x}\\
&= \lim_{x\to 0}\frac{2\cos x - x\sin x}{3\cos x - x\sin x} = \frac{2}{3}
\end{aligned}
$$

References